# MINI ASSESSMENT

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CSE AIML

## API Request Handling

## 1. What does HTTP status code 404 mean, and how should your program handle it?

**Answer:** HTTP status code **404 means "Not Found."** It is returned when the requested webpage, API endpoint, resource, or data cannot be found on the server. For example, if a program sends a request to an incorrect API URL, the server may return a 404 response. The program should check the response status code before using the returned data. If the status code is 404, it should handle the situation gracefully and display a clear message such as "Requested resource not found." The program may also ask the user to verify the URL or the requested resource. Proper handling of 404 errors makes an application easier to use and prevents unexpected failures.

## 2. Why should you use `try-except` when making an API request?

**Answer:** The `try-except` block is an important feature of Python used for **exception handling**. When a program communicates with an API, many problems can occur, such as an unavailable server, network failure, connection timeout, invalid URL, or unexpected response. Without proper error handling, such problems may stop the program and produce an error message. By placing the API request inside a `try` block, the program attempts to perform the operation. If an error occurs, the `except` block catches the error and allows the program to respond safely. For example, it can display "Unable to connect to the server. Please try again later." Therefore, `try-except` improves program reliability, user experience, and stability.

# MINI ASSESSMENT — ANSWERS

## 3. If an API returns an empty JSON response, what should your program do?

**Answer:** If an API returns an empty JSON response, the program should first check whether the response contains the required data before trying to access or process it. An empty response may occur when there are no matching records, the requested information is unavailable, or the server has returned no useful data. The program should not assume that the expected fields or values are always present. Instead, it should perform a condition check and handle the empty result properly. For example, if the API returns an empty list or empty JSON object, the program can display a message such as **"No data found."** This prevents errors such as missing-key or index errors and makes the application more robust. Good API programs should always validate the received data before using it.

## Conclusion

API error handling is an important part of developing reliable applications. HTTP status codes help the program understand whether a request was successful or failed. The `try-except` mechanism helps handle unexpected errors safely, while checking for empty JSON responses prevents the program from processing missing data. Together, these practices make API-based applications more stable, user-friendly, and easier to maintain.

## Key Points:

- 404 indicates that the requested resource was not found.
- `try-except` safely handles API-related exceptions.
- Empty JSON responses should be checked before processing.
- Clear error messages improve the user experience.
- Validating API responses makes programs more reliable.